

SUVAB BARAL

SOFTWARE ENGINEER

CONTACT



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New Jersey, USA

SKILLS

Programming Languages: Node.js, Java, Python, C, C++, C# (.NET framework), React.js, React Native, Angular.

Tools: Linux, Git, Jira, AWS, Azure, Android Studio, Serverless, MQTT, Docker, Kubernetes, Raspberry Pi, Apache Kafka, ORMs, Tensorflow, Keras, CUDA, Bluetooth technology, Postman.

Databases: MSSQL, PostgreSQL, MySQL, AWS Dynamo DB, MongoDB, Redis, InfluxDB.

Languages: English, Nepali, Hindi.

EDUCATION

Stony Brook University, USA

Master of Science (M.S) - Computer Engineering

December 2024

CGPA: 3.83/ 4

Relevant courses:

Machine Learning for Engineering Systems, Computer and Robot Vision, Deep learning algorithms and software, Mobile Sensing Systems and Applications, Parallel processing architectures.

Sri Ramakrishna Engineering College,

Anna University, India

Bachelor of Engineering (B.E) - Computer

Science Engineering

2016 - 2020

CGPA: 9.19 / 10

Relevant courses:

Data Structures, Design and Analysis of Algorithms, Web Technology, Artificial Intelligence and Machine Learning, Cloud Computing.

EXPERTISE

- Skilled in designing robust and scalable full stack software solutions.
- Proficient in machine learning models and Large Language Model (LLM) API integration.
- Over 5 years of experience in designing and developing backend servers and REST APIs using Express.js, Django, Spring, .Net and SQL/NoSQL databases.
- Over 5 years of experience in web and mobile app development with Redux, React, React Native, Angular and Android Studio.
- Experienced in time series analysis and real-time data interaction, specializing in Apache Kafka.
- Proficient with CI/CD tools, including Convex and CircleCI.
- Strong collaboration and leadership as a Scrum Master in an agile software development team, emphasizing continuous and incremental releases.
- Automated unit testing for both frontend and backend applications.

WORK EXPERIENCE

Software Engineer (P2)

10/2024 - Present

.Net Framework, Angular, MSSQL, Azure, Scrum

Trimble Inc., USA

- Designed and developed **end-end solution architecture for Trimble Transportation Maintenance System (TMT)**, covering UI/UX, REST APIs, and database. Key features include Single Sign-On (SSO), Sticky Notes, Reports Dashboard, Inventory Management, and third party integration.
- Enhanced the Trimble Technician Portal (TTP) integrated with TMT, to enable **features such as time tracking, technician management, real-time repair order updates**, and more streamlined field operations.

Research Assistant and Software engineer

08/2023 - 12/2024

Linux, Python, Node.js, MQTT, MySQL, InfluxDB, Raspberry Pi, AWS, Android Studio, Docker

Research Foundation for SUNY, USA

- Developed a scalable and secure **infrastructure for continuous health monitoring system** using Linux, Raspberry Pi, Node.js, Python, MQTT, MySQL, InfluxDB, and AWS IoT for storing and analyzing continuous time series data from Ultra wide-band (UWB) sensors aiding in vital signs monitoring and early detection of symptoms.
- Analyzed various machine learning models, including LLMs and transformers, to optimize activity recognition, location trajectory, and vital signs monitoring.
- Designed a user-friendly **Android application for home self-assessment** of vital signs, body coordination, muscle strength, and more.

Associate Technical Lead

03/2023 - 08/2023

Node.js, React.js, PSQL, AWS, Netsuite 360SB, Docker

Rently, Remote

- Led the strategic design and **collaboration with Netsuite 360SB** to engineer the new Rently MRR (Monthly Recurring Revenue) Billing System.
- Demonstrated a strong commitment to Rently's business vision by enhancing the billing model, and designing the backend for the overall billing experience.
- Achieved a **35% reduction in manual billing workload**, alongside a **30% decrease in billing issues** for the product suite.

Senior Software Engineer

04/2022 - 03/2023

Node.js, Java, PSQL, React Native, AWS DynamoDB, MQTT, Apache Kafka, AWS, Linux Firmware

Rently, Remote

- **Designed IoT device monitoring solution** for smart locks and thermostats using Node.js, Java, AWS EC2, DynamoDB, PostgreSQL, MQTT, and Apache Kafka.
- Developed a **data analysis algorithm** to enhance security, monitoring, and energy conservation from IoT device data.
- Boosted **algorithm performance by 50%** through multithreading and parallel processing.
- Achieved over **70% adoption rate for monitoring software** within Rently's customer base.
- Increased team velocity by 2-4 story points per sprint as a **Scrum Master**, involved during talent recruitment as **technical reviewer, recruiter, and mentor**.

Software Engineer

09/2020 - 03/2022

Node.js, Java, Python, React Native, PSQL, AWS, Docker, Bluetooth, IoT

Rently, Remote

- Mastered full-stack development, creating Rently's cross-platform smart home app in React Native and server-side solutions with Node.js, Java, Python, PostgreSQL, and AWS.
- Led development of the **Bluetooth Low Energy (BLE) library** and **Rently Smart Home Alexa skill** for seamless communication with smart devices.
- Enhanced market presence by leveraging in-house smart locks, eliminating reliance on third-party solutions.
- Maintained over **95% test case coverage** for all projects.

Software Engineer Intern

09/2019 - 08/2020

Node.js, Java, PostgreSQL, AWS, IoT

Rently, India

- Championed the role of Node.js developer, working on four impactful AWS Lambda server projects.
- Developed and maintained features, designed robust **REST APIs, and backend solutions** for the Rently Smart Home mobile app.
- Proficiently utilized AWS services like Lambda, Cognito, DynamoDB, S3, and SQS for server-side and web development.
- Executed agile methodologies, consistently delivering tangible outcomes.